

Growing a Sympoietic Art Organism: An Experiment Tracking the Micro-Dynamics of Cultural Evolution



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Abstract

Observational gaps, survival bias, and noise in retrospective reporting often significantly limit empirical tracking of cultural evolution. Empirical data recording how cultural artifacts, motifs, and lineages form in real time are therefore of great benefit and interest in this area of research. To study these dynamics, we present a closed, participatory modeling experiment implemented through a structured collaborative artmaking exercise. Our experimental design attempts to richly capture the bottom-up emergence of a shared cultural heritage in a small, controlled setting, as well as the influences at play throughout its evolution.

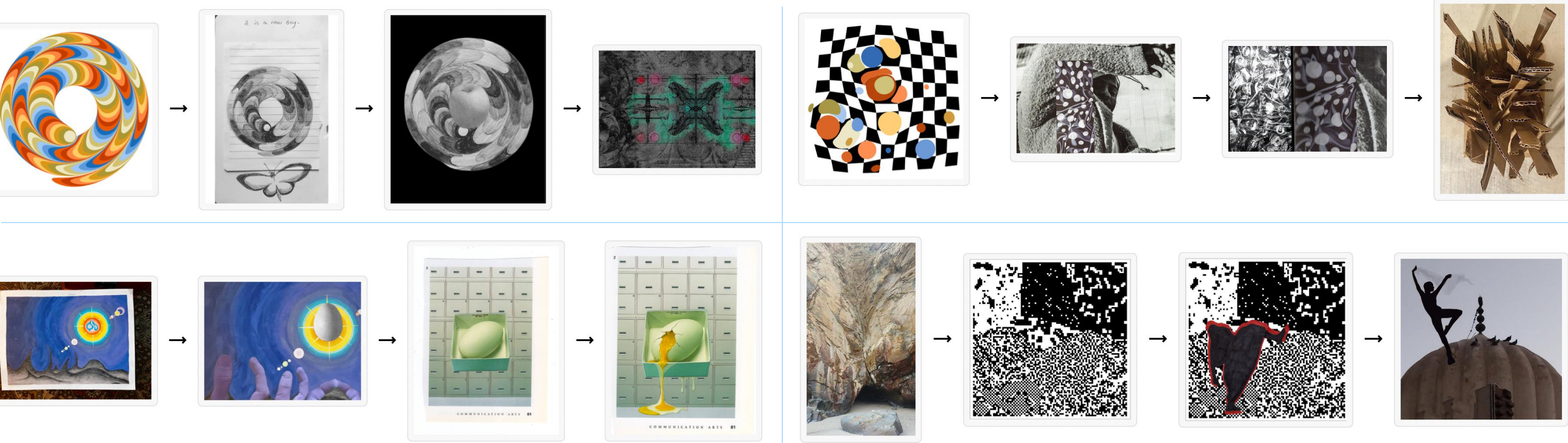
Introduction & Methodology

Prior closed-group experiments have examined individual-level transmission dynamics (who copies whom, what biases shape behavior, when cultural learning is favored [1]) but leave the structural consequences of collective creative decisions largely untracked. Drawing on Haraway's concept of sympoiesis ("making-with") [2], we designed a three-month collaborative artmaking exercise, nicknamed "Sympoietic Art Organism" to study cultural inheritance from a controlled setting in which every artifact and its documented influences were recorded.

- Seeding Phase:** Each participant contributed ≥ 10 artifacts to a shared repository (190 unlinked seed vertices in total). All materials were open to remixing by any participant.
- Co-Creative Cycles:** Over four two-week rounds, participants aimed to contribute ≥ 5 new artifacts citing prior-round works, logged to a shared ledger. Each artifact received a unique ID encoding cycle, artist, and submission index. Citing only prior-round works enforced a strictly feed-forward DAG structure, enabling time-series analysis of network growth.
- Release & Reflection:** Participants reflected on their creative strategies; qualitative accounts complement the quantitative analysis below. All contributions are shared under CC BY-SA.

Through our engagement with each other's works in this micro-cultural setting, we created over 411 original multi-media artifacts connected by 735 verified citation edges. The resulting network provides a detailed, fully auditable record of creative influences throughout the experiment.

Results I: Semantic Phylogeny & Motif Evolution



The DAG encodes a semantic phylogeny of motifs through time. Because citation records and artifacts are preserved together, we can audit where and how influence occurred. Visual inspection of lineages reveals diverse modes of engagement and relationality between pieces:

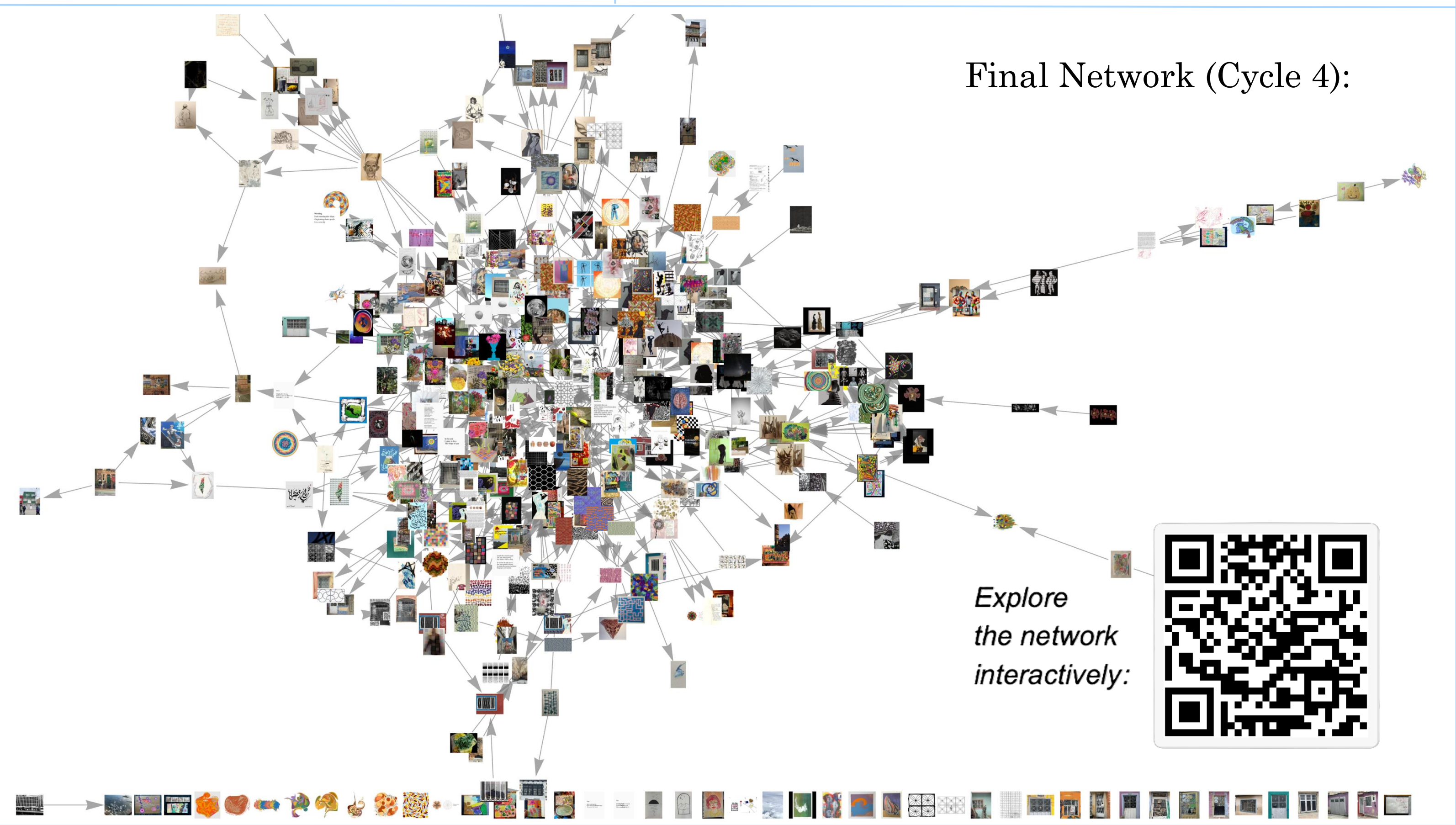
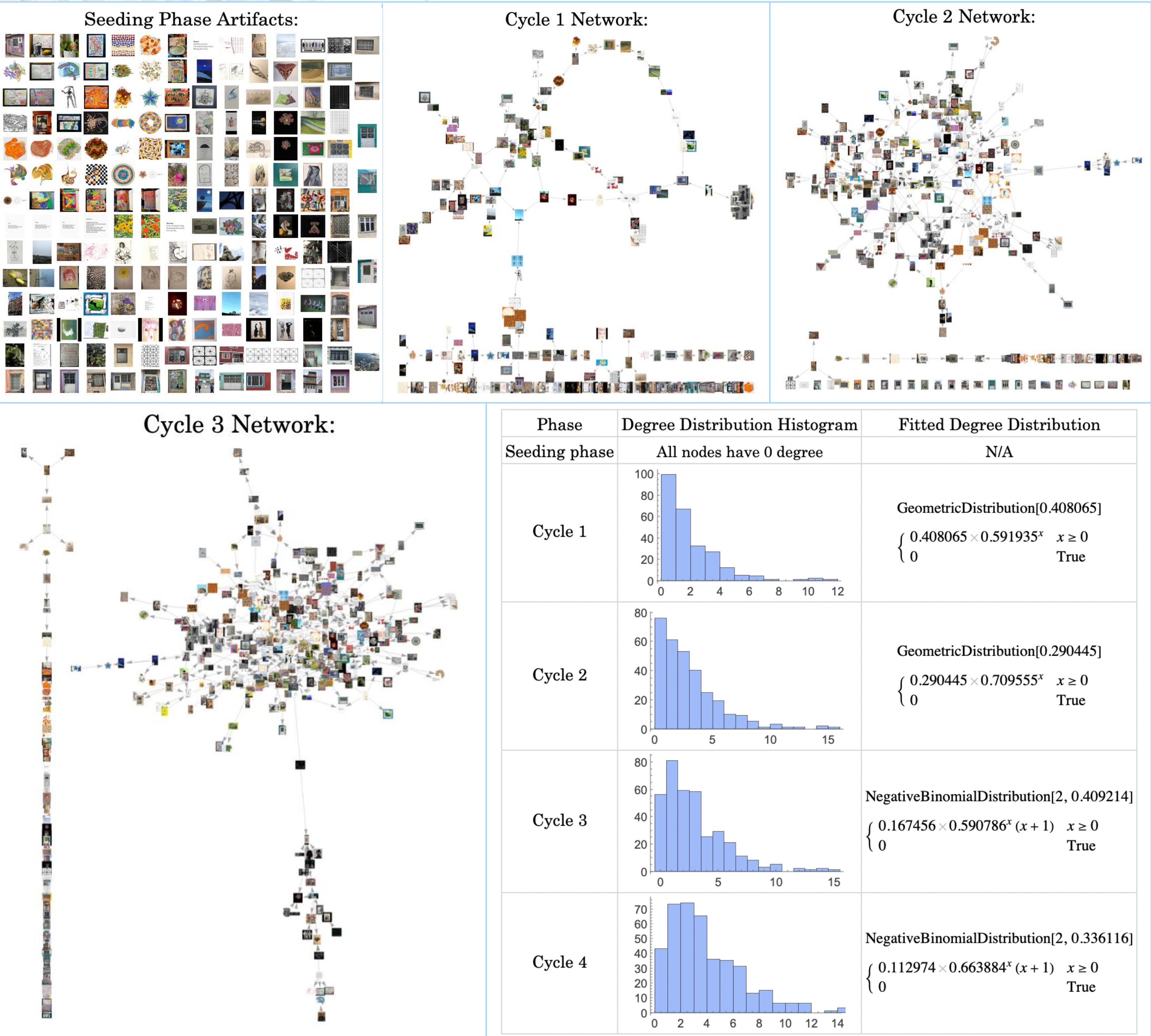
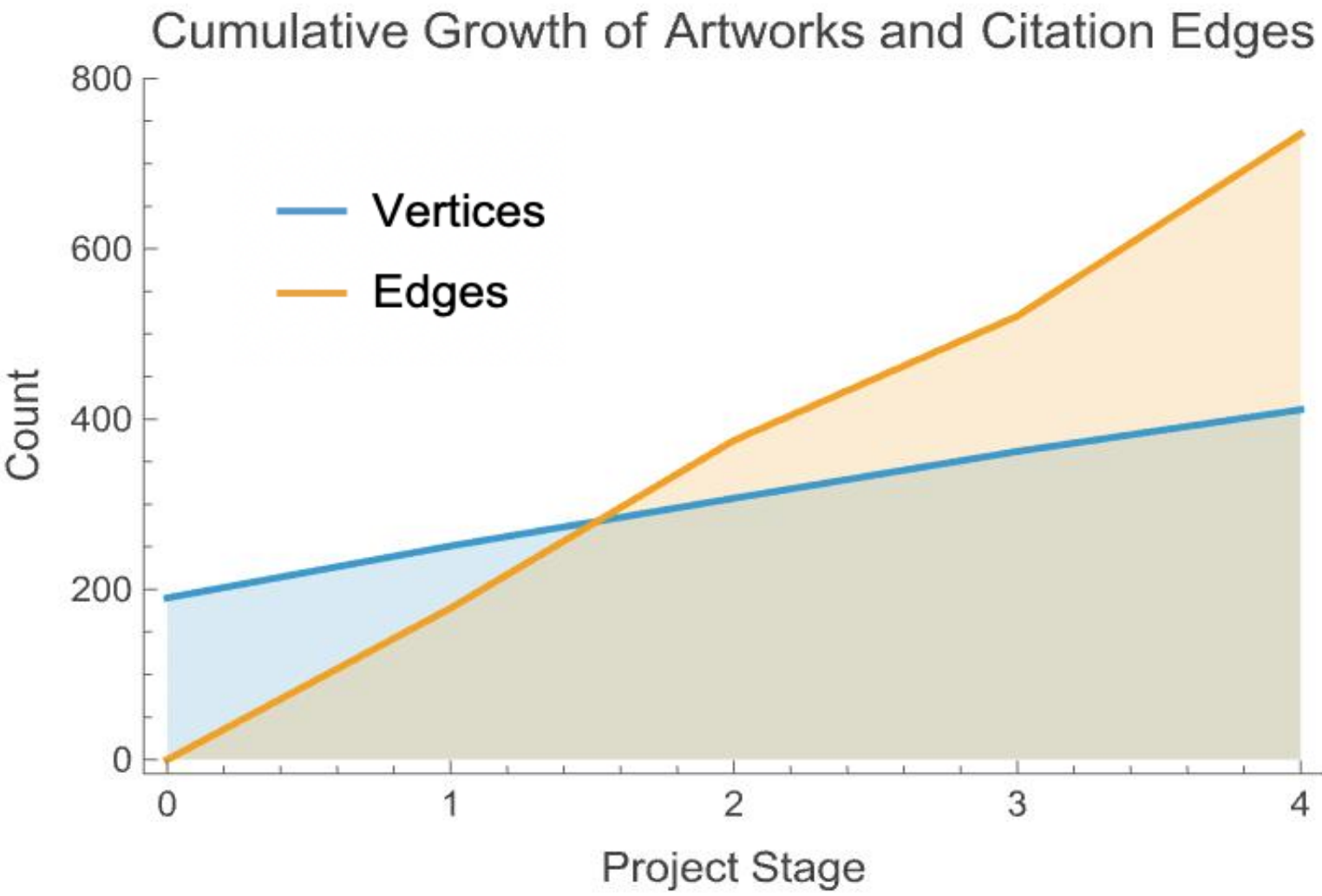
- Inheritance mechanisms:** Lineages diverge in their inheritance mechanisms. Some explicitly reuse and remix specific visual components from their ancestors, while others connect to their source material by analogy or thematic matching, propagating concepts and source elements without copying them directly.
- Lineage forms:** Some lineages naturally have a legible, even narrative progression across generations, while others seem more random, especially when new artifacts subvert their ancestors more radically. The magnitude of the mutations applied to borrowed elements from parent artifacts, their frequency, and the number of their immediate sources all contribute to this apparent randomness. Throughout a lineage, the visual complexity of artefacts may grow or diminish, or both at different times.

Participants reported a tension between wanting to engage with a diverse set of sources and a natural pull toward certain pieces. This suggests a subjective experience of preferential attachment, which we explore in the next section.

Results II: Topological Evolution & Network Dynamics

Cycle 1 produces the first citation edges and a single connected component containing the majority of linked artifacts. Cycle 2 introduces the first triangles in the network; global clustering jumps from 0 to 0.04038. From Cycle 2, citation edge accumulation strictly outpaces vertex addition. In Cycle 3, an elongated tail appears because a single cut-vertex connects the main connected component to an otherwise disconnected subgraph. By Cycle 4, new citation edges bypass the cut-vertex, integrating the cluster into the core. Degree distributions shift from approximately Geometric (Cycles 1–2) to Negative Binomial (Cycles 3–4). While 35% of artifacts receive zero citations, the majority are cited at least once. Yet a few dominant hubs accumulate citations at an accelerating rate [3], generating the heavy-tailed statistical signature of endogenous "rich-get-richer" preferential attachment [4].

Global Network Metrics:					
Project Stage	Vertices	Edges	Density	Global CC	Mean CC
Seeding phase	190	0	0	0	0
Cycle 1	251	178	0.002821	0	0
Cycle 2	307	375	0.003971	0.04038	0.02647
Cycle 3	362	521	0.003971	0.0319	0.02071
Cycle 4	411	735	0.00435	0.03641	0.03973



Conclusion & Recommendations

By treating a collaborative art community as a closed dynamical system, our experiment bridges quantitative network science and qualitative cultural evolution. We observed rapid structural consolidation from isolated seeds into a unified core, driven by cumulative advantage among artifacts. Interestingly, despite this centralization, most nodes ultimately remain structurally peripheral on the main connected component. Alongside this topological maturation, qualitative visual tracing revealed that artifacts relate to their sources in richly varied ways, all of which the experimental design makes visible and auditable.

Unlike post-hoc network reconstruction, our design captures influence as it occurs through a structured exercise in which participants actively document their creative relationships. This makes it possible to capture the structure and qualitative character of creative influence at a level of granularity and detail unavailable in naturalistic settings. We propose this design as a scalable ethnographic tool with particular utility in sociology and digital humanities for tracking cultural transmission dynamics in real time, and in educational settings as both a collaborative creative exercise and a window into the social dynamics of a group.

Acknowledgements

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Citations

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